

Aerodynamic Parameter Tuning

An aero package is not chosen in isolation — adding downforce usually adds drag, and both typically add mass (wing structure, mounts, ballast to rebalance the car). Before committing to a spec, it is worth knowing how much lap time each of these levers is actually worth, and how they trade against one another, rather than tuning downforce, drag, and mass by feel. This report uses the car's lap time simulation to quantify that: mass, downforce coefficient-area (CIA), and drag coefficient-area (CdA) are each swept independently through the full model, and the resulting sensitivities are converted into trade-off ratios that can inform aero package decisions directly.

1. Where Aerodynamics Enters the Lap Time Model

The lap time simulation splits a lap into cornering, where speed is limited by available grip, and straights, where speed is limited by the powertrain and by how late the car can brake. Downforce and drag enter these two problems differently, which is why they end up having very different leverage on lap time.

1.1 Downforce in the Cornering Solve

For each corner, maximum speed is solved from a force balance rather than assumed from a fixed friction circle. At a candidate speed, aerodynamic downforce is computed from CIA and the candidate speed and split across all four contact patches, combined with static load and lateral load transfer to give vertical load at each wheel. Available grip at each wheel comes from a load-sensitive friction model, and the candidate speed is adjusted by a root-finder until the total available lateral force matches the force required to hold that speed through the corner's radius. Because downforce grows with the square of speed, more CIA raises this equilibrium speed at every corner on the track — downforce has a direct, first-order effect on cornering performance.

1.2 Drag in the Straight-Line Simulation

Straights are simulated as a full transient rather than a quasi-steady-state calculation. The car accelerates from the entry speed set by the corner behind it and must brake down to the exit speed required by the corner ahead of it; drag enters both phases as a speed-dependent force, in the forward vehicle model during acceleration and in a distance-domain deceleration ODE during braking. Drag has no direct effect on the cornering solve at all — it is a purely longitudinal force — so its entire cost to lap time is paid on the straights, through reduced acceleration and, partially offsetting, increased braking deceleration.

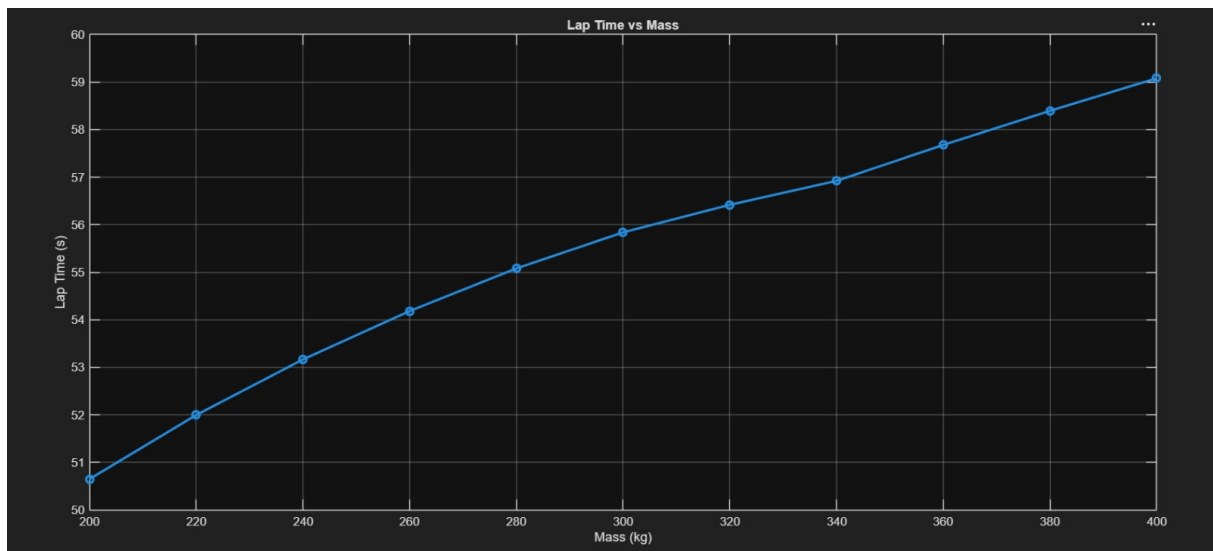
This asymmetry — downforce acting on both cornering and straights, drag acting only on straights — is the reason the two are not equally valuable levers, and motivates quantifying each on its own before trading them against one another.

2. Sensitivity Sweeps

Each parameter below was swept independently through the complete simulation — corner solve and straight-line transient together — holding the other two fixed, so that each curve reflects that parameter's full effect on lap time rather than its effect on one sub-model in isolation.

2.1 Mass

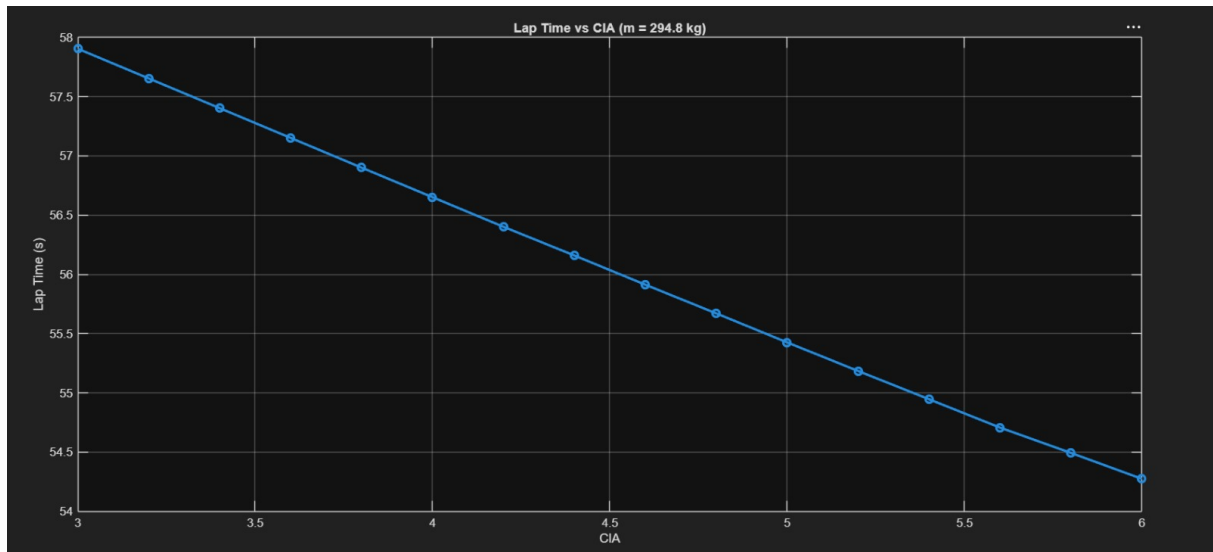
Mass was swept from 200 kg to 400 kg. It enters both the cornering force balance (more mass demands more lateral force to hold the same line, and increases vertical load at each tyre) and the transient acceleration/braking model (more mass to move, per unit of available tractive or braking force). Any change to the aero package that adds structural mass is therefore not free, even before its own drag and downforce are accounted for.



Lap time vs. mass

2.2 Downforce (CIA)

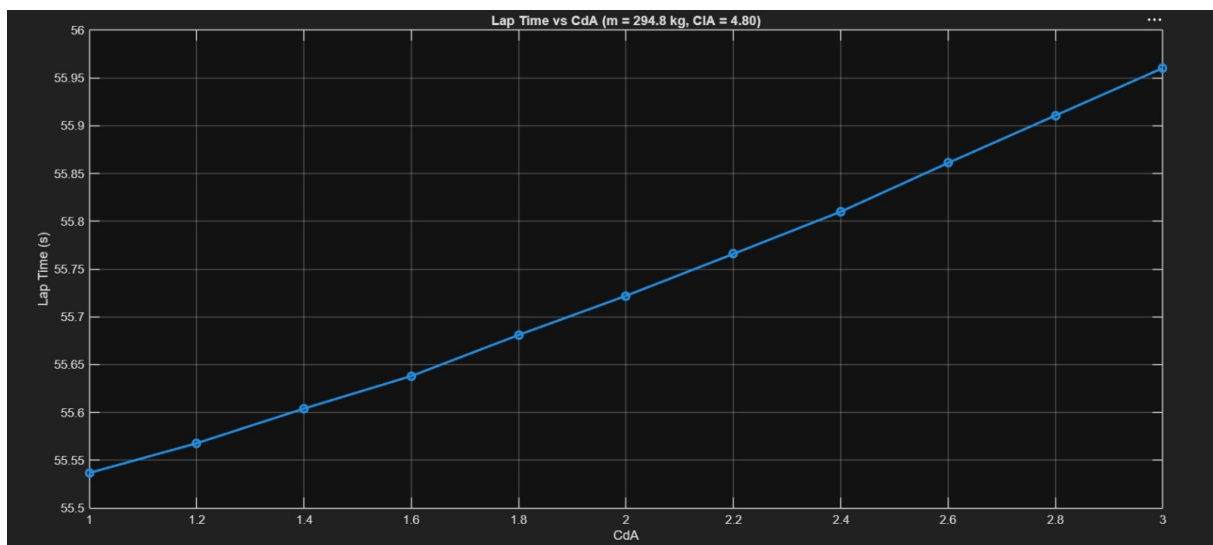
Downforce coefficient-area was swept from 3 to 6, with mass and drag coefficient-area held constant. Consistent with Section 1.1, CIA is the single most effective lever of the three studied — it raises the speed the car can sustain through every corner on the track and also raises the load-dependent traction available during straight-line acceleration.



Lap time vs. downforce coefficient-area

2.3 Drag (CdA)

Drag coefficient-area was swept from 1 to 3, with mass and CIA held constant. As expected from Section 1.2, its cost is paid entirely on the straights, and over the studied range it is a materially smaller lever than downforce.



Lap time vs. drag coefficient-area

3. Trade-Off Ratios

Reading the local slope of each sweep against the others gives the rate at which one parameter can be traded for another while holding lap time constant — how much of a “cost” parameter can be absorbed for a given gain elsewhere before the trade stops being worth it. Two ratios come directly out of the sensitivity results above:

- For every 1 kg of mass added, roughly 0.030 units of CIA need to be added to hold lap time constant — i.e. mass added by the aero package (structure, mounts, ballast) is only “free” if it is buying at least this much downforce.

- For every 1 unit of ClA added, up to roughly 5.6 units of CdA can be added before lap time stops improving — the track tolerates a large drag penalty in exchange for downforce, consistent with a corner-dominated circuit with short straights.

The car's prior aero package ran a ClA/CdA ratio of approximately 3.01. Given the second ratio above — a tolerance of roughly 5.6 units of CdA per unit of ClA at the current operating point — the package sits well inside the drag budget the track can absorb, meaning there was room to trade further drag for downforce before lap time sensitivity to CdA would start to erode the gain.

Taken together, these results identify downforce and mass as the two aerodynamic factors with the greatest leverage on lap time for this car and track, and quantify the terms on which each can be traded against drag and against each other. Tuning the aero package toward higher ClA, within the mass and drag budgets these ratios define, improved lap time over the baseline configuration.